Distance-Constrained Dynamic Subtraction Games

Theorem B: Computer-Assisted Affine Classification

Research Team – draft for independent mathematical review

October 9, 2026

Abstract

Proved separately for 21 fixed parameter values; uniform symbolic proof Working on. This standalone paper gives definitions, proof logic, precise evidence boundaries, and cited sources in the same document.

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1 Theorem A: complete structural results

Proof status: Proved.

1.1 Rules, notation and outcome recurrence

Fix $k \geq 1$, $m \geq 0$ and a heap of $n \geq 0$ counters. From a state (n, s), where $s \in \{1, \ldots, k\}$ is the preceding removal, a legal move removes r counters such that $1 \leq r \leq \min(k, n)$ and $|r - s| > m$. The resulting state is $(n - r, r)$. With no previous removal, the first move has only the first restriction. Normal play applies: a player with no legal move loses. We call losing and winning states respectively $\mathcal{P}$- and $\mathcal{N}$-positions, following standard combinatorial-game terminology [1, 2, 3].

Define the P-state row and its potential P-reaching removals by

$$Q_n = \{s \in \{1, \ldots, k\} : (n, s) \text{ is } \mathcal{P}\}, \tag{1}$$

$$A_n = \{r : 1 \leq r \leq \min(k, n),\ r \in Q_{n-r}\}. \tag{2}$$

In particular $Q_0 = \{1, \ldots, k\}$, since there is no legal removal. Starting position n (no previous move) is a $\mathcal{P}$-position precisely when $A_n = \varnothing$.

Theorem 1.1 (Exact interval theorem)**.** *For every $n \geq 0$, the set Q_n is empty, full, or one interval of consecutive integers. More exactly, if $A_n = \varnothing$, then $Q_n = \{1, \ldots, k\}$. Otherwise*

$$\boxed{Q_n = \left[\max\{1, \max A_n - m\},\ \min\{k, \min A_n + m\}\right] \cap \mathbb{Z}.}$$

An interval whose lower endpoint exceeds its upper endpoint is empty. Any non-full row has at most $2m + 1$ entries.

Proof. The legal move r takes (n, s) to a losing state exactly when $r \in A_n$ and $|r - s| > m$. Hence (n, s) is losing precisely when *every* move in A_n fails its previous-removal constraint. Equivalently, $|r - s| \leq m$ for every $r \in A_n$. If A_n is empty, this holds for all $1 \leq s \leq k$. Otherwise the inequalities $r - m \leq s \leq r + m$ for all $r \in A_n$ intersect to $\max A_n - m \leq s \leq \min A_n + m$, clipped to $1 \leq s \leq k$. Its length is at most $2m + 1$. □

Lemma 1.2 (Finite-memory recurrence)**.** *For fixed k, m, Q_n is computed deterministically from $Q_{n-1}, \ldots, Q_{n-k}$ whenever $n \geq k$.*

Proof. The membership test $r \in Q_{n-r}$ determines A_n for each $1 \leq r \leq k$; then Theorem 1.1 determines Q_n. □

Theorem 1.3 (Eventual P/N periodicity)**.** *For every fixed $k \geq 1$ and $m \geq 0$, the sequence $(Q_n)_{n \geq 0}$ is eventually periodic, and so is the unrestricted starting-position P/N sequence.*

Proof. Each Q_n is a subset of $\{1, \ldots, k\}$, so there are at most 2^k row values. Consider consecutive ordered windows $(Q_n, \ldots, Q_{n+k-1})$. At most 2^{k^2} such windows exist. Theorem 1.2 makes the successor window a deterministic function of its predecessor once the recurrence has entered the full k-move range. Some two windows recur by pigeonhole, and all later windows agree by induction. Thus the rows are ultimately periodic. The starting $\mathcal{P}$ predicate $A_n = \varnothing$ depends on the preceding k rows for $n \geq k$, so it inherits eventual periodicity. □

Corollary 1.4 (Specialization $m = 1$)**.** *Every non-full Q_n has at most three consecutive elements.*

Scope. Eventual periodicity is not a statement about a least period or an affine formula in k. Those stronger claims require separate work.

2 Theorem B: a 21-parameter certified affine regime

Proof status: Proved by exact finite certificates plus induction (computer-assisted).

2.1 Certification lemmas

Lemma 2.1 (Window repetition)**.** *Fix k, m. Suppose for integers $p \geq 1, t \geq 0$, $Q_{t+i} = Q_{t+p+i}$ for every $0 \leq i < k$. Then $Q_n = Q_{n+p}$ for all $n \geq t$.*

Proof. The given k equalities form the induction base. For $n \geq t + k$, each row used in the calculation of A_n is Q_{n-r} for $1 \leq r \leq k$, and the corresponding row for A_{n+p} is Q_{n+p-r}. These rows agree by induction; therefore their potential-removal sets and the resulting Q rows agree by the exact recurrence in Theorem 1.1. Since $n \geq t + k \geq k$, both states allow all removal sizes through k. □

Lemma 2.2 (Starting-position bridge)**.** *Under Lemma 2.1, the starting-position P/N sequence has period p for all $n \geq t + k$. A proposed p-periodic formula is valid for every $n \geq 0$ if it matches the exact starting P/N outcome for $0 \leq n \leq t + k + p - 1$.*

Proof. For $n \geq t + k$, each child row Q_{n-r} agrees with Q_{n+p-r}, and all removals $1 \leq r \leq k$ are available. Thus $A_n = A_{n+p}$, so the starting outcomes agree. The finite bound includes the entire earlier prefix and one complete period beyond $t + k$; any subsequent index reduces to that verified range. □

Theorem 2.3 (Exact certified $A = 86$ regime)**.** *For each integer $58 \leq k \leq 78$, the unrestricted starting position of $D(k, 1)$ with n counters is losing precisely when*

$$\boxed{n \bmod (2k + 88) \in \{0, 86, k + 87\}.}$$

The least positive period of this starting P/N sequence is $2k + 88$.

Finite computation and mathematical induction. Fix one of the 21 values $k = 58, \ldots, 78$. The script `src/verify_theorems_ab.py` computes each Q_n by the integer recurrence of Theorem 1.1. With $p = 2k + 88$, it finds the exact t shown in Table 1 and asserts the k row equalities in Lemma 2.1. It also verifies the proposed start predicate for every $0 \leq n \leq t + k + p - 1$. Lemmas 2.1 and 2.2 therefore prove the infinite-n claim for this fixed k. The script repeats these finite checks for each of the 21 listed integers. It additionally compares the rows and unrestricted starts with an independent direct Sprague–Grundy implementation over the verified range.

To establish minimality, set $R = \{0, 86, k + 87\}$ in the cyclic group of order p. If a nonzero shift fixes a set of three elements, its translation orbits must each have length three, so $R = \{0, p/3, 2p/3\}$. Its smaller nonzero element would therefore be $p/3$. Yet $p/3 = (2k + 88)/3 < 86$ for $58 \leq k \leq 78$, contradicting $\min(R \setminus \{0\}) = 86$. Thus the period is least. □

2.2 Exact reproduced certificates

Table 1: Theorem B, corrected finite-window certificates. The bridge endpoint is inclusive.

k	p	t	bridge end	independent SG
58	204	58	319	PASS
59	206	59	323	PASS
60	208	59	326	PASS
61	210	59	329	PASS
62	212	59	332	PASS
63	214	58	334	PASS
64	216	59	338	PASS
65	218	0	282	PASS

k	p	t	bridge end	independent SG
66	220	0	285	PASS
67	222	0	288	PASS
68	224	0	291	PASS
69	226	0	294	PASS
70	228	0	297	PASS
71	230	0	300	PASS
72	232	0	303	PASS
73	234	0	306	PASS
74	236	0	309	PASS
75	238	0	312	PASS
76	240	0	315	PASS
77	242	0	318	PASS
78	244	0	321	PASS

Unresolved extension (Working on). The 21 instances constitute a completed computer-assisted proof, not a uniform symbolic derivation for arbitrary k. Explicit P-row formulas explaining the numbers 86 and 88 and the boundary at $k = 78$ remain *Working on.* See Theorem C.

Works cited and source verification

References

[1] R. Sprague, "Über mathematische Kampfspiele," *Tôhoku Mathematical Journal*, first series, vol. 41 (1935), pp. 438–444.
Original journal record: https://www.jstage.jst.go.jp/article/tmj1911/41/0/41_0_438/_article. The metadata and page span were checked against the journal listing.

[2] P. M. Grundy, "Mathematics and Games," *Eureka*, no. 2 (1939), pp. 6–8.
Bibliographical entry and link to scanned source at https://oeis.org/A002188/a002188.pdf; the bibliographical description is corroborated by OEIS. The scan was not used for any copied proof.

[3] E. R. Berlekamp, J. H. Conway, and R. K. Guy, *Winning Ways for Your Mathematical Plays*, vol. 1, second ed., A K Peters, 2001.
Publisher record: https://www.routledge.com/Winning-Ways-for-Your-Mathematical-Plays-Volume-1-2-Berlekamp-Conway-Guy/p/book/9781568811307; verified against publisher and author's publication listings.

[4] V. Gurvich, "On Repetition-free Subtraction Games and Vile-dopey Integers," RUTCOR Research Report RRR-01-2012, Rutgers University, January 2012.
Author's publication list: https://rutcor.rutgers.edu/~gurvich/publist.pdf (entry 240). A report transcription was located, but its detailed theorems were *not* invoked here.

Source-use boundary. The P/N and Sprague–Grundy methods are standard background and are credited above. All specific interval, finite-window, template-obstruction and finite-range arguments in

these documents are written self-contained from the game definition, not borrowed from an external proof. Originality relative to *all* prior literature has not been established; a thorough review remains necessary.